

Assessment of Water Quality of Some Selected Boreholes and Hand-Dug Wells in Donga Town of Taraba State, Nigeria

Gamboro Danazumi Maiyaki, Mohammed Bakoji Yusuf, Hajara Garba

Department of Geography, Faculty of Social Sciences, Taraba State University, Jalingo, Nigeria.

*Corresponding author email address: mbyusuff36@gmail.com

Abstract	Article History
Access to safe and sustainable drinking water is essential for achieving sustainable development goal (SDG 6). This study assessed the water quality of selected boreholes and hand-dug wells in Donga town of Taraba State, Nigeria. The study examined physiochemical and biological parameters, calculated WQI value and deduced spatial temporal map of water quality variation. Water samples were collected from 24 points and analyzed using DR 6000 spectrophotometer machine and the result was presented in tables and compared with NSDWQ. Sampled points coordinates were acquired using GPS and was used to generate water quality variation map using inverse distance weighting (IDW) interpolation technique in QGIS 3.44. The findings revealed that, most physiochemical parameters were within permissible limit NSDWQ, except turbidity, nitrate and nitrite. Values ranged between turbidity 10-15NTU, TDS 50.5-90.5ppm, lead concentration 0.01-0.004mg/l, and chloride 1.8-2.8mg/l. However, microbial contaminations was found in all water samples, therefore, the water sources was judged not good for human consumption but good for other domestic use, this necessitate the need for continuous monitoring of groundwater quality in line with SDG 6. The study recommend regular water quality monitoring, protection of groundwater, improved sanitation and waste water management, treatment of contaminated water and promotion of environmental sustainability among others.. Keywords: Water Quality, Selected Boreholes, Hand-Dug Wells, Donga Town , Taraba State	Received: 16 Aug 2026 Accepted: 23 Sept 2026 Published: 30 Sept 2026  Scan QR code to view* License: CC BY 4.0  Open Access article
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1. Introduction

Water was a vital natural resource that was critical for the survival of living organisms and was an essential component of agricultural production, industrial development, sanitation, and socioeconomic activities. Of all the water on Earth, 97% was saline, 3% was freshwater contained in glaciers and polar ice caps, and a significant portion of freshwater was found underground (Benjamin et al., 2021; Joshua et al., 2023; Ma'aruf et al., 2024). Groundwater was therefore an important part of the global freshwater system and provided water for domestic, agricultural, and industrial uses, especially where reliable surface-water supplies and public water infrastructure were limited.

The value of water quality was evidenced in the United Nations Sustainable Development Goals (SDGs), specifically SDG 6, which aimed to ensure the availability and sustainable management of water and sanitation for all. According to the united ions (2023), SDG 6 focused on improving water quality by reducing pollution, eliminating dumping, and significantly increasing the recycling and safe reuse of water in order to improve access to safe and affordable drinking water for all by 2030. Despite these commitments at the global level, access to safe drinking water remained a major problem, especially in

developing countries where insufficient water infrastructure, poor sanitation systems, environmental pollution, and inadequate water-quality monitoring continued to jeopardize water resources (United Nations, 2024).

Human activities and natural processes increasingly degraded groundwater quality. Groundwater systems could be contaminated with physical, chemical, and biological contaminants from agricultural activities, indiscriminate waste disposal, sewage discharge, industrial effluents, poorly constructed sanitation facilities, and other land-use practices. These pressures were further exacerbated in rapidly growing urban communities by high population density and poor environmental stewardship. Waqar et al. (2022) identified water contamination as one of the major environmental concerns globally, while Okorie et al. (2024) highlighted the growing effects of urbanization and industrialization on surface- and groundwater resources and their implications for environmental and public health. Groundwater vulnerability was significant in Nigeria because many groundwater sources used for domestic supply were boreholes and hand-dug wells. Taiwo et al. (2023) reported that anthropogenic activities such as poor sanitation, inappropriate waste disposal, and inappropriate placement of boreholes were increasingly

affecting the safety of groundwater in many communities in Nigeria. A similar situation was reported by Asibor and Ofuya (2019), as the dependence of people on groundwater for domestic and other purposes was observed on a large scale, indicating that groundwater resources needed to be protected from contamination.

When and where water quality declined, it had significant public health consequences. Pathogenic microorganisms and hazardous chemical substances could be transmitted through contaminated water, thereby increasing the likelihood of waterborne diseases and other harmful health effects. The lack of evaluation and monitoring of drinking-water sources was linked to greater susceptibility to waterborne diseases, including cholera, dysentery, and typhoid fever (Joshua et al., 2023a; Nesto et al., 2024). Unsafe water and lack of sanitation were also linked to significant morbidity and mortality globally by Abera (2020). These concerns were especially alarming in communities that relied directly on untreated groundwater for drinking water and other household needs. Therefore, periodic water-quality monitoring was required to detect potential contaminants, determine the appropriateness of water for specific uses, and guide proper management of water resources and public health practices (Babuji et al., 2023). The monitoring of water quality was especially important because it generated data that helped to assess whether water was of a quality suitable for domestic purposes, including human consumption, and to determine whether it met the required quality standards.

The problem of groundwater protection was compounded by the lack of public water-supply facilities in Nigeria. The poor quality and reliability of piped-water supplies had led many households, especially those in rural and peri-urban areas, to rely on boreholes and hand-dug wells as primary sources of water for household use. Groundwater was typically considered less susceptible than surface water to contamination, but its quality could deteriorate when water sources were poorly located, poorly constructed, poorly maintained, or situated close to pollution sources. Sources that could contaminate groundwater included agricultural activities, indiscriminate disposal of solid waste, sewage infiltration, inadequate management of domestic wastewater, proximity to pit latrines, and inadequate protection of wellheads. Densely populated areas where sanitation services and water points were located close to one another posed particular concerns.

Taraba State was not spared from these problems. Available evidence indicated that groundwater contamination was a serious problem in some parts of the state, partly due to poor sanitation and waste-management practices that contributed to the deterioration of groundwater sources such as boreholes. Omale et al. (2024) noted high levels of contamination in borehole water sources in Taraba State and attributed the contamination to poor sanitation. Their findings clearly showed the need for location-specific assessment because groundwater quality could vary significantly depending on local geology, land use, settlement patterns, sanitation practices, waste disposal, and the construction and maintenance of individual water sources.

Donga Town, being an important settlement in Donga Local Government Area of Taraba State, experienced increasing water demands due to domestic and socioeconomic activities. Boreholes and hand-dug wells were important sources of water for the residents because there was no adequate and reliable centralized water-supply system. However, the quality of groundwater could not be assumed based on the groundwater source alone. Groundwater in the town could have been affected by sanitation practices, waste disposal, land use, source protection, and the proximity of wells and boreholes to potential contamination sources. Therefore, it was important to characterize these conditions in order to assess the quality of groundwater for domestic consumption and identify areas that could benefit from treatment.

Although available literature reported on groundwater quality in many parts of Nigeria and Taraba State, little information was available regarding the combined physicochemical, bacteriological, and spatial characteristics of groundwater from boreholes and hand-dug wells in Donga Town in particular. This information gap prevented relevant authorities and communities from identifying water sources that might have been unsafe, prioritizing interventions, and developing appropriate groundwater-protection measures. Therefore, a systematic assessment was conducted to determine the quality status of groundwater sources, assess their suitability for drinking, and identify spatial variations in water quality within the town.

In this context, the present study evaluated selected physicochemical and bacteriological characteristics of water from boreholes and hand-dug wells in Donga Town, Taraba State, Nigeria. In particular, the measured water-quality parameters were compared with the World Health Organization (WHO) guideline values; the Water Quality Index (WQI) was calculated to ascertain the overall water-quality condition of the sampled water resources; and the spatial variability of selected water-quality parameters was analyzed using GIS-based spatial interpolation. The vision study is to provide empirical data for groundwater-quality monitoring, environmental sanitation planning, sustainable water-resources management, and public-health protection in Donga Town, while also contributing to the achievement of global SDG 6.3 on water quality and pollution reduction.

2. Materials and Methods

2.1. The Study Area

The study was conducted in Donga Town, the administrative headquarters of Donga Local Government Area (LGA) in Taraba State, Nigeria. Donga LGA is situated in the southern part of Taraba State within the Middle Benue region. The town and its surrounding communities constitute an important settlement area in the LGA and depend substantially on groundwater sources, particularly boreholes and hand-dug wells, for domestic water supply. Geographically, the study area lies approximately between latitude 7°43'00" N and longitude 10°03'00" E. The area falls within the tropical environment of southern Taraba, where seasonal rainfall, settlement expansion, agricultural activities, and associated land-use practices may influence groundwater conditions.

Administratively, Donga LGA shares boundaries with Kurmi Local Government Area to the northeast, Bali Local Government Area to the northwest, Wukari Local Government Area to the southwest, and Takum Local Government Area to the southeast. The LGA is administratively divided into 10 political wards, namely Asibity, Fada, Gyata-Aure, Akate, Kadarko, Suntai, Marraraba, Gayama, Kumbo, and Nyita. These administrative divisions provide the spatial framework within which settlements, agricultural activities, sanitation facilities, and groundwater sources are distributed.

According to 2006 population census of the National Population Commission (NPC), Donga town had an estimated population (projected to 2025) of 134,525 persons in 2025. The population and settlement structure create substantial demand for water for domestic, agricultural, and socioeconomic activities. In the absence of adequate and reliable public water-supply infrastructure, groundwater

obtained through boreholes and hand-dug wells constitutes an important source of water for residents. Consequently, the quality and safety of these groundwater sources are of considerable environmental and public-health importance.

The physical and socioeconomic characteristics of the study area provide an important context for assessing groundwater quality. Human activities, including domestic waste disposal, sanitation practices, agricultural activities, and the location and management of water facilities, may influence the physicochemical and bacteriological characteristics of groundwater. The assessment of water quality in Donga Town is therefore necessary to determine the suitability of groundwater for domestic use and to identify spatial variations that may require targeted environmental-management interventions. The geographical location and administrative setting of the study area are presented in Figure 2.

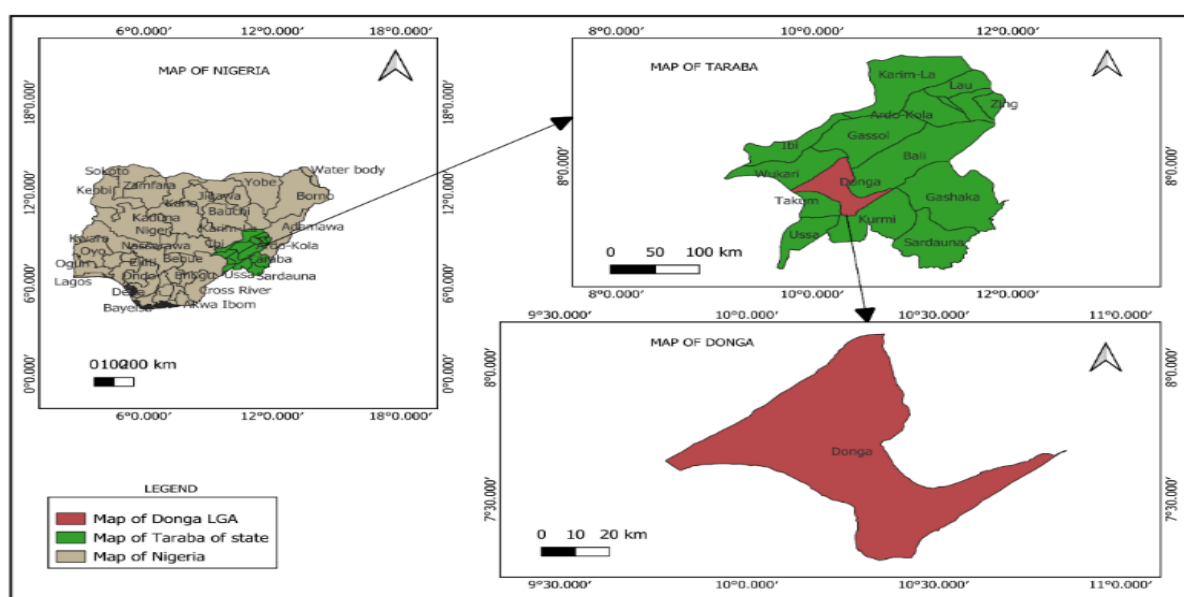


Figure 2: Map of the study area

Source: Adapted from administrative map of Nigeria

2.2. Methods

The study adopted a descriptive and experimental research design to assess the physicochemical and bacteriological quality of groundwater obtained from selected boreholes and hand-dug wells in Donga Town, Taraba State, Nigeria. The descriptive component was used to characterize the existing quality conditions of the groundwater sources, while the experimental component involved laboratory analysis of water samples collected from the selected sampling locations. The study integrated laboratory-based water-quality assessment with geospatial techniques to examine the spatial distribution and variation of selected water-quality parameters across the study area.

The study population comprised all identified boreholes and hand-dug wells within Donga Town. A field enumeration conducted for the study identified approximately 350 boreholes and 800 hand-dug wells, giving a total sampling frame of 1,150 groundwater sources. Four political wards—Akate, Asibiti, Fada, and Gyata-Aure—were selected because they fall within the geographical extent of Donga Town. A probability sampling procedure was

adopted, with simple random sampling used to select the groundwater sources for laboratory analysis. A total of 24 groundwater samples, comprising 12 borehole-water samples and 12 hand-dug-well samples, were collected from the selected sampling locations.

To enable identification and spatial analysis, the sampling points were assigned unique codes according to their locations and source types. The borehole sampling points were coded as AKB1, AKB2, AKB3, AB1, AB2, AB3, GB1, GB2, GB3, and FB1, FB2, FB3, while the corresponding hand-dug-well sampling points were coded as AKW1, AKW2, AKW3, AW1, AW2, AW3, GW1, GW2, GW3, FW1, FW2, and FW3. The geographic coordinates of each sampling location were recorded using a Global Positioning System (GPS) device. These coordinates provided the spatial reference for mapping and interpolation of the measured water-quality parameters.

Water samples were collected from the selected boreholes and hand-dug wells using appropriate sampling procedures and subsequently transported for laboratory analysis. Selected

physicochemical and bacteriological parameters were determined in the laboratory, with the physicochemical analyses involving the use of a DR 6000 spectrophotometer for parameters requiring spectrophotometric determination. The analytical results were subsequently compared with relevant Nigeria standard of drinking water quality (NSDWQ) drinking-water guideline values to assess the suitability of the sampled groundwater for human consumption.

The laboratory results were organized and summarized using tables, frequencies, ranges, and other appropriate descriptive statistics. The Water Quality Index (WQI) was calculated using the weighted arithmetic index method originally developed by Brown et al. (1972). The WQI provided an integrated assessment of overall water quality by combining the measured parameters according to their relative importance and comparison with the corresponding drinking-water standards.

The general procedure involves assigning a unit weight to each parameter based on its recommended standard value and subsequently calculating the quality rating and weighted sub-index for each parameter.

Weighted Arithmetic Index formula by (Brown et al., 1972)

Step 1: calculate the weighted unit (W_n) factors for each parameters

$$W_n = \frac{K}{S_n} \quad \text{Where:}$$

W_n = unit weight of n^{th} parameter

$$K = \frac{1}{S_1} + \frac{1}{S_2} + \dots + \frac{1}{S_3} \quad \text{for all selected parameters}$$

S_n = standard permissible value of the n^{th} parameters

Step 2: calculate sub index (Q_n) value

$$Q_n = \frac{(V_n - V_o)}{(S_n - V_o)} * 100 \quad \text{Where:}$$

Q_n = sub index or quality rating of the n^{th} parameter

V_n = the observed/ measured value of the n^{th} parameter in water sample

V_o = Ideal value of the n^{th} parameters in pure water. For most parameters, $V_o = 0$, except for PH which is equal to 7.0

Step 3: calculate the WQI

$$WQI = \sum W_n Q_n / \sum W_n$$

Step 4: Categories of the water quality status

0-25 excellent, 26-50 good, 51-75 poor, 76-100 very poor, and greater than 100 unfit for consumption.

For spatial analysis, the GPS coordinates of the sampling points were imported into QGIS 3.44. The Inverse Distance Weighting (IDW) interpolation technique was applied to estimate the spatial distribution of selected water-quality parameters across un-sampled locations. IDW was selected because it estimates values at un-sampled locations based on the values of surrounding sampled points, with closer observations receiving greater influence than more distant observations. The resulting interpolated surfaces were used to generate spatial water-quality variation maps and identify areas of relatively higher or lower concentrations of the assessed parameters.

3. Result and Discussion

3.1 Laboratory Analysis of Physiochemical and Biological Parameters

The result of the laboratory analysis of the physiochemical and bacteriological parameters, calculated WQI table and spatial variation map is presented in this section (Tables 1 and 2).

Table 1: Results of Physiochemical and Bacteriological Analysis of Boreholes and Hand-Dug Wells, Akate and Asibiti Wards, Donga LGA

S/N	Parameter	Akb1	Akb2	Akb3	Akw1	Akw2	Akw3	NSDWQ	Status
Akate Ward									
1	Turbidity (NTU)	10	12	15	15	13	15	5 (NTU)	not accepted
2	TDS (ppm)	60.5	5.0	8.50	85.5	90.50	60.5	500PPM	accepted
3	PH	7.50	8.10	8.50	7.80	7.90	8.20	6.5-8.5	accepted
4	Hardness	80	90	110	170	110	140	150	Accepted
5	Lead (PB)	0.001	0.0003	0.001	0.01	0.001	0.0001	0.01mg/l	Accepted
6	Chloride (cl)	2.80	2.30	1.80	3.10	2.20	2.10	250mg/l	Accepted
7	Nitrate (NO_3)	57	62	60	53	51	55	50mg/l	Accepted
8	Nitrite (NO_2)	1.0	0.4	2.1	0.6	1.5	1.2	0.2mg/l	Accepted
9	Iron	0.002	0.03	1.02	0.001	0.02	0.001	0.3mg/l	Accepted
10	Fecal coliform	1	1	2	2	1	2	0 CFU/100mg/l	Not accepted
11	Escherichia coli	1	1	1	2	1	1	0 CFU/100mg/l	Not accepted
12	Salmonella	1	2	1	1	1	2	0 CFU/100mg/l	Not accepted
Asibiti Ward									
		Ab1	Ab2	Ab3	Aw1	Aw2	Aw3	NSDWQ	Status
1	Turbidity (NTU)	15	14	15	10	12	10	5	Not accepted
2	TDS (ppm)	74	75	70	50.5	60	50.5	500	Accepted
3	PH	80.5	80.50	8.55	8.50	8.00	8.50	6.5-8.5	Accepted
4	Hardness	85	90	90	120	210	95	150	2 points not accepted
5	Lead (PB)	0.003	0.002	90	120	210	95	150	Accepted
6	Chloride (cl)	2.50	2.40	2.20	2.70	2.50	2.50	250mg/l	Accepted
7	Nitrate (NO_3)	51	54	51	53	52	51	50mg/l	Not accepted
8	Nitrite (NO_2)	0.4	1.1	0.2	2.1	0.6	0.3	0.2mg/l	Not accepted
9	Iron	0.003	0.01	0.005	0.004	0.002	0.001	0.3mg/l	Accepted
10	Fecal coliform	1	1	2	2	2	2	0CFU/100mg/l	Accepted
11	Escherichia coli	2	1	1	1	2	1	0CFU/100mg/l	Accepted
12	Salmonella	0.3	0.2	0.2	0.4	1	0.4	0CFU/100mg/l	Accepted

Source: Researcher's field work, (2026)

Table 2: Results of Physiochemical and Bacteriological Analysis of Boreholes and Hand-Dug Wells, Gyata-aure and Fada Wards, Donga LGA

S/N	Parameter	Gb1	Gb2	Gb3	Gw1	Gw2	Gw3	NSDWQ	Status
Gyata-aure Ward									
1	Turbidity (NTU)	14	12	10	13	14	15	5 NTU	Not Accepted
2	TDS (ppm)	90.0	60.5	60.5	90	80.5	70.5	500 PPM	Accepted
3	PH	7.80	8.00	7.80	7.90	7.90	7.90	6.6-8.5	Accepted
4	Hardness	90	10	70	110	114	80	150mg/l	Accepted
5	Lead (PB)	0.003	0.003	0.001	0.004	0.0002	0.01	0.01mg/l	Accepted
6	Chloride (cl)	2.10	2.20	2.10	2.20	2.40	2.40	250mg/l	Accepted
7	Nitrate (NO ₃)	51	52	52	54	51	53	50mg/l	Not Accepted
8	Nitrite (NO ₂)	0.4	0.5	0.3	1	0.4	0.4	0.2mg/l	Not Accepted
9	Iron	0.001	0.002	0.01	0.003	0.001	0.003	0.3mg/l	accepted
10	Fecal coliform	1	2	01	2	1	1	0 CFU/00ml	Not accepted
11	Escherichia coli	1	1	0.1	1	0.1	1	0 CFU/00ml	Not accepted
12	Salmonella	1	1	0.2	0.1	1	1	0 CFU/00ml	Not accepted
Fada Ward									
		Fb1	Fb2	Fb3	Fw1	Fw2	Fw3	NSDWQ	Status
1	Turbidity (NTU)	15	13	12	15	15	11	5 NTU	Not accepted
2	TDS (ppm)	70.5	65.5	50.5	85.5	70	70.5	500 PPM	Accepted
3	PH	8.00	7.50	8.50	8.50	7.80	70.90	6.5-8.5	Accepted
4	Hardness	110	90	110	120	110	130	150mg/l	Accepted
5	Lead (PB)	0.001	0.002	0.001	0.001	0.001	0.002	0.01mg/l	Accepted
6	Chloride (cl)	2.10	1.90	2.50	2.20	2.10	2.50	250mg/l	Accepted
7	Nitrate (NO ₃)	53	52	51	52	52	51	50mg/l	Not Accepted
8	Nitrite (NO ₂)	1	0.4	2	0.6	0.4	0.3	0.2mg/l	Not Accepted
9	Iron	0.002	0.001	0.01	0.2	0.003	0.001	0.3mg/l	Accepted
10	Fecal coliform	1	1	1	1	1	2	0 CFU/00ml	Not accepted
11	Escherichia coli	1	1	1	1	1	2	0 CFU/00ml	Not accepted
12	Salmonella	1	1	1	1	1	1	0 CFU/00ml	Not accepted

Source: Researcher's Field Work, (2026)

The laboratory results presented in Tables 1 and 2 indicate that the turbidity values across all sampled locations ranged from 10 to 15 NTU. These values were above the Nigeria standard of drinking water quality (NSDWQ) permissible limit of 5 NTU, indicating that the sampled groundwater visibly cloudy and contained high concentrations of suspended particles. This finding align with the observations of Ma'aruf et al. (2024), who reported elevated turbidity and alkalinity levels and emphasized the need for appropriate water treatment interventions to address associated health concerns. The relatively low turbidity observed in the present study may be attributed to the natural filtration of groundwater as it percolates through soil and geological layers during recharge.

The total dissolved solids (TDS) concentrations ranged from 50 to 90.5 mg/L, substantially below the NSDWQ guideline value of 500 mg/L. The low TDS concentrations indicate relatively low levels of dissolved salts, minerals, and other dissolved substances in the groundwater. This suggests that the sampled groundwater sources are relatively fresh and chemically suitable for domestic purposes. Similar findings were reported by Jagaba et al. (2020), who observed that the physicochemical parameters investigated in their study were generally within acceptable WHO limits. The relatively favorable TDS levels observed in the present study are therefore indicative of good groundwater quality and contribute to the broader objectives of Sustainable

Development Goal 6 (SDG 6), which seeks to ensure the availability and sustainable management of water and sanitation for all.

The pH values ranged from 7.50 to 8.50, indicating that the groundwater was predominantly neutral to slightly alkaline. These values fall within the NSDWQ acceptable range of 6.5–8.5. The hardness values, however, varied considerably, ranging from 70 to 210 mg/L. Most of the sampled water sources recorded hardness values below the NSDWQ guideline value of 150 mg/L, although Akw1 (170 mg/L) and Aw2 (210 mg/L) exceeded the recommended limit. The relatively high hardness observed in these samples may be associated with the dissolution of calcium- and magnesium-bearing minerals and rocks within the groundwater system (Babuji et al., 2023). Although water hardness at the levels observed does not generally constitute a major direct health risk, excessive hardness may have adverse effects on domestic activities, including washing, cooking, and the performance of household appliances.

Lead (PB) concentrations were generally low across the sampled locations, ranging from 0.0001 to 0.01 mg/L. These values were either below or equal to the NSDWQ guideline value of 0.01 mg/L, suggesting that lead contamination was not widespread in the study area. The relatively low concentrations may be associated with the limited presence of

major industrial activities and restricted discharge of hazardous wastes into the groundwater system. Mehmet et al. (2024) reported that industrial emissions and associated anthropogenic activities are important sources of contamination in groundwater and river systems. The present finding is also consistent with Tukura et al. (2022), who reported relatively low concentrations of heavy metals in water and sediments of the River Donga, Taraba State.

Similarly, chloride concentrations across all sampled locations ranged from 1.80 to 3.10 mg/L, which is substantially below the NSDWQ guideline value of 250 mg/L. This finding is consistent with Aremu et al. (2017), who reported relatively low concentrations of selected contaminants in groundwater and river-water samples from Wukari, Taraba State. The low chloride concentrations recorded in the present study suggest limited influence from sewage discharge, industrial effluents, or saline intrusion. Consequently, chloride does not appear to constitute a significant chemical threat to the domestic suitability of the groundwater sources investigated (WHO, 2024).

In contrast, nitrate (NO_3) concentrations were elevated in all the sampled water sources, with values exceeding the NSDWQ guideline value of 50 mg/L. This finding indicates significant nitrate contamination of the groundwater and represents an important water-quality concern in the study area. Elevated nitrate concentrations in groundwater are commonly associated with anthropogenic activities, particularly the excessive application of inorganic fertilizers, poor waste-management practices, leakage from septic systems, and infiltration of animal wastes into the subsurface environment (Al-Taai, 2021). Prolonged exposure to excessive nitrate concentrations in drinking water may pose health risks, particularly to infants, due to the potential development of methemoglobinemia (WHO, 2024; Al-Taai, 2021). The elevated nitrate levels observed in the present study therefore suggest that agricultural practices and inadequate sanitation may be contributing substantially to groundwater contamination. If left unaddressed, this condition could undermine progress towards SDG 6 by compromising access to safe and sustainable drinking-water resources.

Similarly, nitrite (NO_2) concentrations were elevated in all the sampled locations. The elevated nitrite concentrations indicate contamination associated with organic wastes, sewage infiltration, and microbial decomposition processes. The occurrence of relatively high nitrite levels, particularly in shallow groundwater sources, provides further evidence of the influence of inadequate sanitation and surrounding environmental pollution on groundwater quality (Rehman et al., 2019). The findings therefore highlight the need for improved sanitation, appropriate waste disposal, and regular monitoring of groundwater sources within the study area.

Iron concentrations were generally low across the sampled locations and remained below the NSDWQ guideline value of 0.3 mg/L. This finding suggests that iron contamination does not constitute a major water-quality problem in the study area. Low iron concentrations are also desirable from an aesthetic and domestic-use perspective because excessive iron can impart undesirable taste and colour to water and may

contribute to staining and the accumulation of deposits in household plumbing and equipment (WHO, 2024).

The bacteriological analysis revealed the presence of faecal coliforms, *Escherichia coli*, and *Salmonella* in all sampled water sources. The detection of these organisms is of considerable public-health significance because indicator organisms such as faecal coliforms and *E. coli* should be absent from drinking water intended for human consumption. Their presence provides evidence of faecal contamination of the groundwater sources. Such contamination may result from inadequate sanitation, open defecation, the proximity of wells to refuse dumps and septic tanks, and the infiltration of contaminated surface runoff into shallow groundwater systems (Al-Taai, 2021). The occurrence of these potentially pathogenic microorganisms indicates that the sampled groundwater sources may not be safe for direct consumption without appropriate treatment. Their presence also increases the potential risk of waterborne diseases, including diarrheal diseases, cholera, typhoid fever, and dysentery (Rehman et al., 2019).

The findings further indicate that hand-dug wells were generally more contaminated bacteriologically than boreholes. This pattern may be explained by the shallow depth and greater exposure of hand-dug wells to surface-derived contaminants. In contrast, boreholes typically access deeper groundwater and may therefore be relatively better protected from direct contamination, although their safety ultimately depends on construction quality, sanitary protection, and local hydrogeological conditions. Similar observations were reported by Akoachere et al. (2008) and Jagaba et al. (2020), who found that shallow groundwater sources are generally more vulnerable to contamination than deeper aquifers because of their limited depth and susceptibility to poorly managed surrounding land uses and inappropriate siting of water facilities. The greater vulnerability of hand-dug wells observed in the present study underscores the importance of proper well siting, sanitary protection, regular water-quality monitoring, and appropriate treatment before consumption.

3.2 Water Quality Index and Status

The summary sheet of WQI for the 24 points analyzed and the water status indicate that, WQI values ranged from approximately 2.13 to 38.83 (See table 3).

Based on the adopted Water Quality Index (WQI) classification scheme, water with WQI values of 0–25 is classified as **excellent**, 26–50 as **good**, 51–75 as **poor**, and 76–100 as **very poor**. The results presented in the table indicate that the majority of the sampled locations recorded WQI values below 25, corresponding to excellent water quality. These locations include AKB1, AKB2, AKB3, AKW2, AKW3, AB2, AW3, GB3, GW2, FB1, FB2, FB3, FW1, FW2, and FW3. The predominance of excellent WQI values suggests that groundwater quality at most sampling points is generally favorable and suitable for domestic purposes based on the parameters incorporated into the WQI calculation.

In contrast, seven sampling locations—AB1, AW1, AW2, AW3, GB1, GB2, and GW1—recorded WQI values ranging from approximately 29 to 39, placing them within the **good** water-quality category. Although these values indicate a relatively lower quality than that observed at the excellent sites, the groundwater

at these locations remains within the good classification under the adopted WQI scheme. Notably, AKW1 and GW3 recorded WQI values of 96.7, indicating **very poor water quality**. These locations therefore represent priority areas requiring further investigation to identify the specific parameters responsible for the elevated WQI values and to determine appropriate management and treatment measures.

The application of WQI provides an integrated assessment of groundwater quality by combining several physicochemical and microbiological parameters into a single numerical value, thereby facilitating the interpretation and comparison of overall water quality among sampling locations. Overall, the WQI results indicate that groundwater quality across the study area is predominantly **excellent**, with 14 of the 24 sampling locations

falling within this category. Seven locations were classified as **good**, while two locations were classified as **very poor**. The predominance of excellent and good WQI classifications suggests that groundwater quality is generally favorable across the study area. However, the occurrence of very poor water quality at AKW1 and GW3 indicates localized deterioration and highlights the need for targeted monitoring and investigation of potential contamination sources. Thus, although groundwater quality is generally favorable, localized anthropogenic activities and site-specific hydrogeological conditions may be influencing water quality at particular sampling points.

Table 3: Coordinates, WQI and Water Status

S/N	Site	Latitude	Longitude	WQI Value	Status
1	AKB 1	7.723774	10.026006	9.8	Excellent
2	AKB2	7.745605	10.028792	19.8	Excellent
3	AKB3	7.728271	10.020227	20.7	Excellent
4	AKW1	7.749298	10.022742	96.7	Very poor
5	AKW2	7.729811	10.023419	10	Excellent
6	AKW3	7.724965	10.014537	1.2	Excellent
7	AB1	7.727591	10.051081	29.2	Good
8	AB2	7.723951	10.069505	19.6	Excellent
9	AB3	7.758836	10.059399	38.9	Good
10	AW1	7.726595	10.058955	38.83	Good
11	AW2	7.723989	10.059981	29.12	Good
12	AW3	7.718152	10.068299	19.5	Excellent
13	GB1	7.704982	10.042207	29.1	Good
14	GB2	7.716478	10.044973	29.12	Good
15	GB3	7.719957	10.049893	9.9	Excellent
16	GW1	7.721433	10.04173	38.8	Good
17	GW2	7.702659	10.047455	2.1	Excellent
18	GW3	7.712191	10.040965	96.7	Very poor
19	FB1	7.744915	10.054742	9.82	Excellent
20	FB2	7.735418	10.045374	19.42	Excellent
21	FB3	7.744377	10.042286	9.93	Excellent
22	FW1	7.748516	10.045872	11.1	Excellent
23	FW2	7.739899	10.048878	9.82	Excellent
24	FW3	7.731379	10.04268	19.44	Excellent

Source: Researcher's field work, (2026)

3.3 Spatial Variation of Groundwater Quality

The spatial distribution of groundwater quality, as represented by the WQI values, reveals a heterogeneous pattern across the study area. Although most sampling locations exhibit excellent groundwater quality, variations in WQI values indicate localized differences in groundwater conditions. These variations may be associated with differences in land use, proximity to potential sources of contamination, well depth, well construction, sanitation conditions, and the underlying hydrogeological characteristics of the area.

The spatial distribution of the WQI values is presented in Figure 3. Borehole sampling locations are represented by **red dots**, whereas hand-dug well sampling locations are represented by **yellow dots**. The map provides a visual representation of the geographical distribution of groundwater quality and facilitates the identification of areas with relatively favorable or deteriorated water-quality conditions. The spatial pattern is particularly useful for identifying localized areas requiring further investigation and for supporting groundwater management and protection strategies.

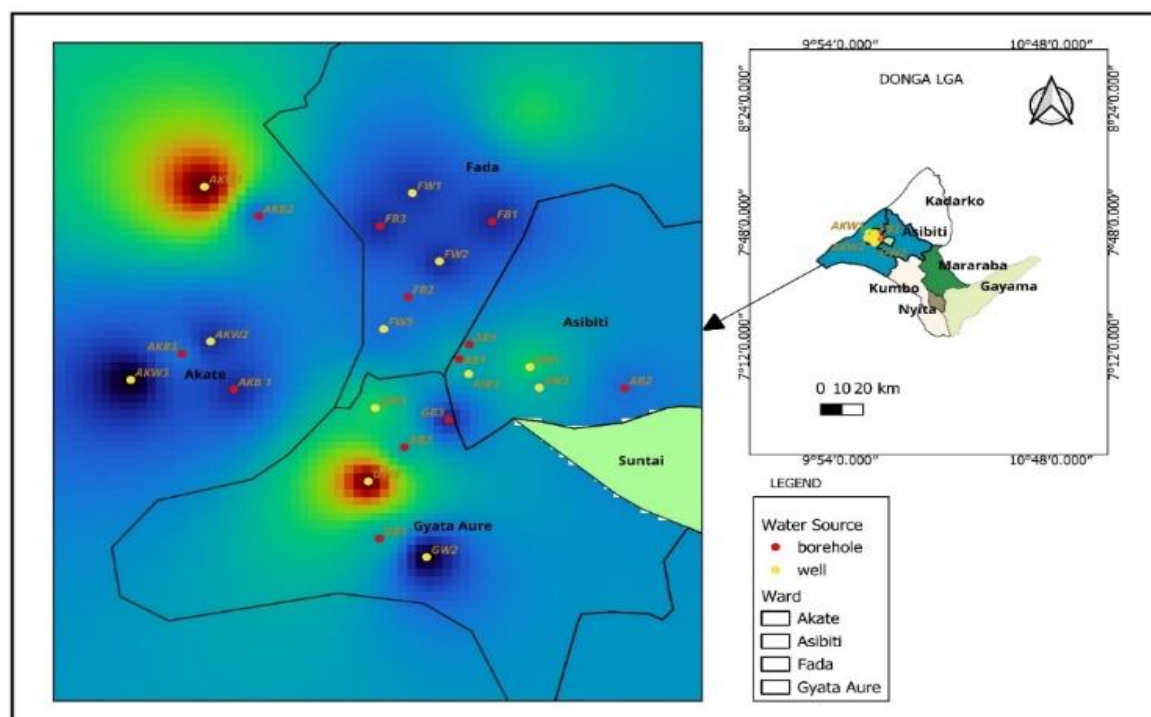


Figure 3: Spatial variation of groundwater quality in selected wards of Donga LGA
Source: Researcher's field work, (2026)

The WQI map illustrates a distinct spatial variation in groundwater quality, represented by a colour gradient ranging from blue through green, yellow, and orange to red. The blue areas represent relatively low WQI values and, consequently, better groundwater quality, whereas the progression from green and yellow to orange and red indicates increasing WQI values and a corresponding deterioration in groundwater quality. The most pronounced high-WQI hotspot, represented by orange and red zones, occurs around Akate, with another distinct orange-red hotspot observed around Gyata Aure. In contrast, extensive blue zones are evident around Akate, Fada, and Asibiti, indicating comparatively low WQI values and favorable groundwater quality. Green and yellow zones occur between these areas, representing transitional groundwater-quality conditions.

The spatial distribution of WQI values reveals considerable heterogeneity in groundwater quality across the study area, with values ranging from 1.2 to 96.7. Despite this spatial variability, the overall pattern indicates that groundwater quality is predominantly excellent to good across most of the study area. The majority of sampling locations recorded relatively low WQI values, ranging from 1.2 to approximately 29.12, suggesting that the groundwater remains generally suitable for domestic use based on the parameters incorporated into the WQI assessment. The limited occurrence of elevated WQI values further suggests that groundwater contamination is localized rather than widespread. Similar spatial patterns have been reported in GIS-based groundwater-quality assessments, in which the majority of sampling locations exhibited good to excellent water quality, while only a small proportion showed localized deterioration (Kawo & Karuppanan, 2020).

The predominance of low WQI values may reflect favorable hydro geochemical conditions and relatively limited anthropogenic pressure in substantial portions of the study area. Groundwater quality is often influenced by land-use activities, waste disposal practices, agricultural intensity, and the nature of interactions between groundwater and the surrounding geological materials. Areas experiencing lower anthropogenic pollutant loading may consequently maintain relatively favorable groundwater conditions (Singh et al., 2021). The extensive low-WQI zones identified in the study area may therefore indicate comparatively stable groundwater conditions and limited accumulation of dissolved contaminants.

The occurrence of very low WQI values around parts of Akate, Fada, and Asibiti indicates areas of excellent groundwater quality. These zones may be associated with favorable recharge conditions and dilution processes that reduce the concentration of dissolved constituents in groundwater. Recharge-dominated groundwater systems can exhibit relatively lower ionic concentrations where rapid infiltration and replenishment limit the accumulation of dissolved solutes and, in some settings, reduce the duration of water-rock interaction (Kawo & Karuppanan, 2020; Adimalla & Qian, 2022). However, confirmation of this interpretation would require supporting hydrogeological and hydro chemical evidence, particularly information on groundwater recharge rates, aquifer characteristics, and groundwater flow patterns.

Conversely, moderate to high WQI values, particularly those recorded around Akate (AKW1) and Gyata Aure (GW3), indicate localized deterioration in groundwater quality. While the moderate WQI values do not necessarily indicate severe contamination, the elevated values at these locations suggest higher concentrations of one or more water-quality parameters

contributing to the overall index. Potential sources may include agricultural runoff, leakage from septic systems, inappropriate waste disposal, and other localized anthropogenic activities. Similar relationships between anthropogenic activities and localized deterioration of groundwater quality have been documented in previous groundwater studies (Kawo & Karuppannan, 2020). Nevertheless, identifying the specific sources responsible for the elevated WQI values requires further investigation using source-apportionment techniques, land-use analysis, and detailed hydro chemical assessment.

Overall, the WQI map demonstrates that groundwater quality in the study area is predominantly excellent to good, with localized areas of relatively poor water quality. The generally favorable WQI values indicate that groundwater remains suitable for domestic use at most sampling locations based on the parameters assessed. However, the localized high-WQI zones, particularly around AKW1 and GW3, warrant continued monitoring and further investigation. Regular groundwater-quality assessment, improved sanitary protection of water sources, appropriate waste-management practices, and careful control of potentially contaminating land-use activities are recommended to prevent further deterioration and maintain the quality of groundwater resources in the study area.

4. Conclusion and Recommendation

4.1 Conclusion

This study assessed the physicochemical and bacteriological quality of groundwater from boreholes and hand-dug wells in selected wards of Donga Local Government Area (LGA), Taraba State. A total of 24 groundwater samples were collected and analyzed for 12 water-quality parameters: turbidity, total dissolved solids (TDS), pH, hardness, lead (PB), chloride (CL), nitrate (NO_3), nitrite (NO_2), iron (Fe), faecal coliforms, *Escherichia coli* (*E. coli*), and *Salmonella*. The laboratory results were evaluated against the relevant World Health Organization (WHO) guidelines and the Nigerian Standard for Drinking Water Quality (NSDWQ).

The findings indicate that most of the physicochemical parameters, including TDS, pH, chloride, iron, and lead, were within the applicable WHO guideline values across the sampled groundwater sources. However, elevated turbidity, nitrate and nitrite concentrations were recorded at several locations, while the bacteriological analysis revealed the presence of faecal coliforms, *E. coli*, and *Salmonella*. These findings indicate that, despite generally favorable physicochemical characteristics, microbial contamination and elevated nitrogen compounds constitute important constraints to the suitability of the groundwater for direct human consumption.

Consequently, the groundwater sources cannot be considered suitable for drinking without appropriate treatment and effective sanitary protection. However, based on the parameters assessed, the water may remain suitable for certain non-potable domestic uses, subject to appropriate management and use-specific water-quality requirements. The Water Quality Index (WQI) assessment further demonstrated that groundwater quality was predominantly excellent to good

across the study area, although localized areas of very poor quality were identified. These localized deterioration zones highlight the need for targeted monitoring and investigation of potential contamination sources.

Overall, the study demonstrates the importance of continuous groundwater-quality monitoring in Donga LGA. Maintaining the quality of groundwater sources with favorable conditions while addressing identified contamination problems is essential for protecting public health and supporting progress towards Sustainable Development Goal 6 (SDG 6), particularly the target of improving access to safe and sustainable water resources.

4.2 Recommendations

The study therefore recommended that.

- i. Regular Water-Quality Monitoring and Treatment: Relevant government agencies and local authorities should establish regular monitoring programmes for boreholes and hand-dug wells in Donga Town. Monitoring should cover physicochemical and bacteriological parameters, and the results should be compared with the Nigerian Standard for Drinking Water Quality (NSDWQ) and applicable World Health Organization (WHO) guidelines. Water sources that show bacteriological contamination or elevated nitrate and nitrite concentrations should be appropriately treated before being used for drinking and other domestic purposes.
- ii. Improved Sanitation, Waste Management, and Protection of Water Sources: Sanitation and waste-management practices around boreholes and hand-dug wells should be improved to reduce groundwater contamination. Pit latrines, septic tanks, soak away systems, refuse dumps, and other potential contamination sources should be properly sited away from groundwater sources. Boreholes and hand-dug wells should also be properly constructed, covered, sealed, and maintained, with appropriate sanitary aprons and drainage structures to prevent the entry of contaminated surface water.
- iii. Public Health Education and Sustainable Agricultural Practices: Residents of Donga Town should be sensitized to the health risks associated with consuming untreated groundwater and encouraged to practice appropriate household water treatment, hygienic water handling, and safe storage. Farmers should also be encouraged to adopt environmentally sustainable agricultural practices, including judicious fertilizer application, proper use of organic fertilizers, integrated pest management, and reduced dependence on indiscriminate pesticide application to minimize agricultural contamination of groundwater.
- iv. GIS-Based Groundwater Protection and Investigation of Contamination Hotspots: Areas identified through the Water Quality Index (WQI) and spatial analysis as having relatively poor water quality should be subjected to further hydrogeological and land-use investigations. Geographic Information Systems (GIS) should be incorporated into groundwater

monitoring to identify vulnerable areas, map contamination patterns, guide the siting of future boreholes and wells, and support location-specific groundwater protection and environmental sanitation planning.

- v. Community Participation and Groundwater-Quality Management: Sustainable groundwater management in Donga Town should involve local communities, water-user groups, local government authorities, health agencies, environmental agencies, and agricultural extension officers. Communities should participate in the protection, maintenance, and monitoring of groundwater sources and should promptly report activities that may threaten groundwater quality. Relevant authorities should also maintain a groundwater-quality database to support long-term monitoring, planning, and public-health protection

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